

Trench Fever (*Bartonella quintana* infection): A Comprehensive Disease Characteristics Report

Disease: Trench Fever · **MONDO ID:** MONDO:0005991 · **Category:** Infectious Disease **Causal agent:** *Bartonella quintana* (α-proteobacteria; formerly *Rochalimaea quintana*) **ICD-10:** A79.0 · **ICD-11:** 1C30.0 · **MeSH:** D014205 (Trench Fever) Evidence base: aggregated disease-level literature (reviews, case-control studies, one randomized trial, experimental microbiology). No individual EHR or genomic data files were provided; this report synthesizes primary literature retrieved via PubMed (27 papers reviewed, 11 findings confirmed).

Summary

Trench fever is a **louse-borne bacterial infection** caused by *Bartonella quintana*, a small, fastidious, facultatively intracellular α-proteobacterium that is **restricted to a single natural reservoir — humans — and a single principal vector, the human body louse (*Pediculus humanus humanus*)**. First characterized in 1915 among soldiers in the trenches of World War I (hence the name), the disease has re-emerged over the past three decades as "**urban trench fever**" among homeless and socially marginalized populations in high-income cities in the US and Europe ([PMID: 16494745](#)). It is fundamentally a **disease of poverty and body-louse infestation**, not a genetic disease; there are no known human causal genes, susceptibility loci, or heritable risk factors.

Clinically, *B. quintana* produces a **broad spectrum** running from the classic self-limited relapsing febrile illness ("five-day fever," with headache and severe shin/tibial bone pain) through **chronic afebrile bacteremia, blood-culture-negative endocarditis, chronic lymphadenopathy**, and — particularly in immunocompromised hosts such as people with HIV — the **vasoproliferative disorders bacillary angiomatosis and peliosis** ([PMID: 16494745](#); [PMID: 15891120](#)). Two mechanistic axes dominate the pathophysiology: (1) the laterally acquired **Trw type IV secretion system (T4SS)** mediates host-specific adhesion to and invasion of erythrocytes, producing the hallmark long-lasting intraerythrocytic bacteremia; and (2) a suite of factors — the **Vomps** trimeric autotransporter adhesins, the **VirB/VirD4 T4SS**

with its Bep effectors, and the BafA autotransporter (a VEGF analog) — converge on HIF-1/VEGF signaling to drive endothelial proliferation, tube formation, and angiogenesis (PMID: 20548954; PMID: 32678094; PMID: 21536788; PMID: 23163798).

Diagnosis rests on serology, blood culture (often prolonged), and molecular detection (PCR of *gltA/ftsZ*/16S rRNA). The best-supported treatment for chronic bacteremia is **doxycycline plus gentamicin**, validated by a randomized open trial (eradication in 7/7 per-protocol treated vs 2/9 controls, P=0.003) (PMID: 12821469) and a systematic review/meta-analysis (PMID: 23602630). Prevention is fundamentally **body-louse control and improved hygiene/social conditions**; there is no vaccine. This report synthesizes 11 confirmed findings across all 15 template sections.

Key Findings

Finding 1 — Trench fever is caused by *Bartonella quintana*, a human-restricted, louse-borne pathogen

Bartonella quintana is an α -proteobacterium (formerly *Rochalimaea quintana*) whose lifestyle is **restricted to human hosts and louse vectors**. It was the first *Bartonella* species definitively linked to human disease and was characterized as the agent of trench fever, a disease "described in 1915 on the basis of natural and experimental infections in soldiers" (PMID: 16494745). After decades of relative obscurity, it re-emerged as a pathogen of the **urban homeless**: in a study of homeless persons, *B. quintana* was isolated from blood culture in 5.3% of patients (PMID: 15643300).

The verbatim evidence: "*Bartonella quintana*, a pathogen that is restricted to human hosts and louse vectors, was first characterized as the agent of trench fever. The disease was described in 1915 on the basis of natural and experimental infections in soldiers" (PMID: 16494745).

Key identifiers: MONDO:0005991; ICD-10 A79.0; ICD-11 1C30.0; MeSH D014205. This is an **infectious, aggregated disease-level entity** — knowledge is derived from case series, cohort studies, microbiology, and molecular biology, not from Mendelian/EHR genetic resources (no OMIM/Orphanet gene entry applies because it is not heritable). Common synonyms: five-day fever, quintan fever, His-Werner disease, shin bone fever, wolhynia fever, urban trench fever.

Finding 2 — *B. quintana* causes a clinical spectrum from classic trench fever to chronic bacteremia, endocarditis, and bacillary angiomatosis

B. quintana is "responsible for a wide spectrum of conditions, including chronic bacteremia, endocarditis, and bacillary angiomatosis" (PMID: 16494745). The classic acute syndrome is a relapsing fever recurring at ~5-day intervals ("quintana" = fifth), with **headache, dizziness, and characteristic severe pain in the shins/tibiae**. Beyond the acute illness, chronic afebrile bacteremia can persist for months; endocarditis is an important severe outcome. In a series from Sfax, Tunisia, *B. quintana* endocarditis **"represents 9.8% of all endocarditis"** in that setting, with high IgG titers ($\geq 1:800$) supporting the diagnosis (PMID: 15891120). Chronic infection can also present as isolated lymphadenopathy/adenomegaly, including in seronegative patients (PMID: 8727894). In immunocompromised hosts (notably HIV), the organism drives **vasoproliferative** disease — bacillary angiomatosis and peliosis.

Clinical syndrome	Host context	Key features	HPO/phenotype terms
Classic trench fever	Immunocompetent	Relapsing 5-day fever, headache, severe shin/bone pain	Fever HP:0001945; Headache HP:0002315; Bone pain HP:0002653
Chronic afebrile bacteremia	Homeless, louse-exposed	Persistent intraerythrocytic bacteremia	Bacteremia HP:0031864
Blood-culture-negative endocarditis	Valvulopathy/chronic infection	Vegetations, high IgG titers; may need valve surgery	Endocarditis HP:0100584
Chronic lymphadenopathy	Sometimes seronegative	Adenomegaly	Lymphadenopathy HP:0002716
Bacillary angiomatosis / peliosis	Immunocompromised (HIV)	Vasoproliferative skin/bone/visceral lesions	Cutaneous vascular lesions

Finding 3 — Trw T4SS mediates erythrocyte invasion; BafA drives VEGF-dependent angiogenesis

Two experimentally demonstrated virulence mechanisms anchor the pathophysiology.

(1) Erythrocyte invasion via the Trw T4SS. Using signature-tagged mutagenesis in surrogate *Bartonella* species, the **Trw type IV secretion system** was shown to be directly involved in erythrocyte adhesion; its genes **"encode components of the type IV secretion system (T4SS) Trw, demonstrating that this virulence factor laterally acquired by the Bartonella lineage is directly involved in adherence to erythrocytes"** (PMID: 20548954). This host-specific adhesion enables the hallmark **long-lasting intraerythrocytic bacteremia**.

(2) VEGF-driven vasoproliferation via BafA. The *Bartonella* autotransporter **BafA acts as a VEGF analog: "BafA interacts with vascular endothelial growth factor (VEGF) receptor-2 and activates the downstream signaling pathway, suggesting that BafA functions as a VEGF analog. A BafA homolog from a related pathogen, Bartonella quintana, is also functional"** (PMID: 32678094). This explains the vasoproliferative lesions (bacillary angiomatosis) directly at the molecular level.

Findings 4 & 10 — Treatment: doxycycline + gentamicin eradicates chronic bacteremia (randomized + meta-analytic evidence)

The best-supported regimen for **chronic *B. quintana* bacteremia** is **doxycycline plus gentamicin**. A **randomized open trial** (Foucault, Raoult, Brouqui 2003) in homeless patients with blood-culture-positive *B. quintana* compared gentamicin 3 mg/kg/day IV for 14 days + doxycycline 200 mg/day orally for 28 days versus no treatment. **"Intention-to-treat analysis of 20 included patients showed eradication of bacteremia in 7 out of 9 treated patients versus 2 out of 11 untreated controls (P = 0.01). In the per-protocol analysis, eradication was obtained for 7 out of 7 treated patients versus 2 out of 9 untreated controls (P = 0.003)"** (PMID: 12821469).

A **systematic review and meta-analysis** of human bartonellosis treatment corroborates this: **"In chronic bacteremia, gentamicin and doxycycline significantly increased the resolution rate"** (PMID: 23602630) — though the authors note the evidence base is limited and dominated by observational studies at high risk of bias. Expert-guideline regimens (Rolain et al. 2004) extend this: **endocarditis** typically requires prolonged therapy (often doxycycline + an aminoglycoside) and may require **valve surgery**; **bacillary angiomatosis** responds to macrolides (erythromycin) or doxycycline given for months. NCIT term suggestions: Doxycycline (NCIT:C692), Gentamicin (NCIT:C575), Erythromycin (NCIT:C608).

Finding 5 — *B. quintana* has a reduced ~1.58 Mb genome shaped by secondary genome reduction; T4SS drives host adaptation

Comparative genomics of the genus shows *Bartonella* are α -proteobacteria with **host-restricted lifestyles characterized by long-lasting intraerythrocytic infection in a specific mammalian reservoir and arthropod transmission**. Among sequenced genomes (circular chromosomes 1.44–2.62 Mb; 1,283–2,136 genes; a syntenic core of 959 genes), *B. quintana* underwent "**massive secondary genome reduction**" — it has the **smallest genome (~1.58 Mb, ~1,300 genes)** — consistent with adaptation to a single human host plus louse vector (PMID: 19696500). The **type IV secretion systems (Trw, VirB/VirD4)** were laterally acquired and are "**type IV secretion systems that adopted prominent roles in host adaptation and specificity**" (PMID: 19696500). This reductive evolution reflects the pathogen's dependence on host-derived metabolites (host-integrated metabolism) and its narrow ecological niche.

Finding 6 — Bacillary angiomatosis is epidemiologically linked to homelessness, low income, and lice, with subcutaneous and lytic bone lesions

A case-control study of 49 HIV-associated bacillary angiomatosis-peliosis patients versus 96 matched controls found that 23/49 (47%) were *B. quintana* and 26/49 (53%) *B. henselae*. Crucially, tissue tropism differed by species: "**Subcutaneous and lytic bone lesions were strongly associated with *B. quintana*, whereas peliosis hepatis was associated exclusively with *B. henselae***" (PMID: 9407154). Risk factors also segregated: patients "**with *B. quintana* were clustered and were characterized by low income (P=0.003), homelessness (P = 0.004), and exposure to lice (P= 0.03)**", whereas *B. henselae* was tied to cat/flea exposure (PMID: 9407154). This confirms both the **skin/bone tissue tropism** of *B. quintana* and its **socioeconomic risk profile**.

Finding 7 — Transmission is via infectious body-lice feces (horizontal); no true transovarial (vertical) transmission

Humans acquire *B. quintana* through **contaminated body-lice feces**, not the louse bite directly: "**Horizontal transmission from the body louse vector (*Pediculus humanus humanus*) to a human host occurs through contact with infectious louse feces containing a high concentration of the bacteria**" — typically inoculated into skin abrasions or bite-scratch excoriations (PMID: 35803580). Controlled experiments testing vertical transmission found bacterial DNA on egg surfaces (fecal contamination of the egg sheath) but no viable

organisms internally: **"viable *B. quintana* could not be cultured from the hemolymph of adult female lice or from within eggs that were surface sterilized, indicating a lack of true transovarial transmission"** (PMID: 35803580). Vertical transfer therefore **"probably has a limited impact on the dynamics of transmission to humans."** This has direct public-health implications: interrupting the human↔louse cycle (delousing) breaks transmission.

Finding 8 — Vmps adhesins and the VirB/D4 T4SS–Bep effector system converge on HIF-1/VEGF to drive vasoproliferation

B. quintana expresses **variably expressed outer membrane proteins (Vmps)**, which are **trimeric autotransporter adhesins (TAAs)** analogous to *B. henselae*'s **Bartonella adhesin A (BadA)**. In a study of adherence under static and bloodstream-like dynamic flow, investigators **"analyzed three different TAAs (Bartonella adhesin A [BadA] of Bartonella henselae, variably expressed outer membrane proteins [Vmps] of Bartonella quintana, and Yersinia adhesin A [YadA] of Yersinia enterocolitica) for mediating bacterial adherence to ECM and endothelial cells"** (PMID: 21536788) — establishing Vmps as *B. quintana*'s ECM/endothelial adhesins (binding fibronectin, collagen).

Mechanistically, in the closely related *B. henselae* the pathway is well defined: **"Expression of Bartonella adhesin A (BadA) is crucial for bacterial autoagglutination, adhesion to host cells, binding to extracellular matrix proteins and proangiogenic reprogramming via activation of hypoxia inducible factor (HIF)-1"** (PMID: 18627378). In parallel, the VirB/VirD4 T4SS delivers effector proteins into endothelial cells: **"VirB/D4 translocates several Bartonella effector proteins (Beps) into the cytoplasm of infected ECs, resulting, e.g. in uptake of bacterial aggregates via the invasome structure, inhibition of apoptosis and activation of a proangiogenic phenotype"** (PMID: 23163798). Notably, BadA and VirB/D4 are frequently mutually exclusive during in vitro passage — a phase-variation phenomenon relevant to culture-dependent diagnostics.

Finding 9 — Humans are the primary reservoir, but *B. quintana* DNA is increasingly detected in animals; dogs model *Bartonella* endocarditis

The natural reservoir of *B. quintana* is **humans (NCBI Taxon 9606)**, transmitted by the body louse. However, molecular surveys increasingly detect *B. quintana* DNA in **non-human hosts**. A survey of pet cats in Urmia, Iran found **"15 % of the cats (30 out of 200 blood samples) tested positive for the *B. quintana* gene, with a 95 % confidence interval of 10.71 % to 20.61 %"** (100% sequence identity to the reference) (PMID: 38199070). For comparative disease modeling, **dogs** develop naturally occurring *Bartonella* endocarditis, making **"canids... the**

most interesting naturally occurring animal model for the human disease" (PMID: 12860639). Experimental mechanistic work uses surrogate rodent-adapted species — "mouse-specific *Bartonella birtlesii*, human-specific *Bartonella quintana*, cat-specific *Bartonella henselae* and rat-specific *Bartonella tribocorum*" in adhesion/invasion assays (PMID: 20548954) — and the body louse itself serves as an in-vivo vector model.

Finding 11 — Epidemiology: ~5% of sheltered homeless bacteremic; rising; worldwide wherever body lice occur

In a 4-year study of 930 homeless people in Marseilles, lice were found in 22%, and *B. quintana* was isolated from blood culture in 50 (5.3%); critically, "the number of bacteremic patient increased from 3.4% to 8.4% ($p = 0.02$) over the 4 years of the study" — evidence of a rising trend (PMID: 15643300). Geographically, the pathogen is cosmopolitan wherever body-lice infestation occurs: a molecular survey of body lice detected "the presence of *B. quintana* in lice collected from all locations except the Congo" — spanning France, Russia, Peru, Zimbabwe, and Burundi (PMID: 9986818). *B. quintana* is one of only three body-lice-borne human pathogens, alongside *Rickettsia prowazekii* (epidemic typhus) and *Borrelia recurrentis* (louse-borne relapsing fever); co-exposure occurs, with documented *R. prowazekii* seroconversion in a *B. quintana*-bacteremic homeless person (PMID: 15891141).

Mechanistic Model / Interpretation

Ordered causal chain: from louse feces to clinical disease

1. Body louse (*Pediculus humanus humanus*) feeds on a bacteremic human
| leads to
2. *B. quintana* replicates in the louse gut and is shed in LOUSE FECES
(high bacterial concentration; NOT transovarial – no vertical spread)
| results in
3. Contaminated feces inoculated into skin abrasions / bite-scratch
excoriations of a new human host (horizontal transmission)
| leads to
4. Bacteria seed a primary niche (endothelial cells / dermal tissue)
| then (via Trw T4SS)
5. Trw type IV secretion system mediates HOST-SPECIFIC ADHESION to and
INVASION of erythrocytes
| results in
6. Long-lasting INTRAERYTHROCYTIC BACTEREMIA (immune-privileged niche;
enables relapsing fever + chronic bacteremia + louse re-acquisition)
 - BRANCH A (acute/chronic systemic disease):
relapsing 5-day fever, headache, severe shin/bone pain;
chronic afebrile bacteremia → endocarditis (valve
vegetations, culture-negative)
 - BRANCH B (vasoproliferative disease, esp. immunocompromised):
Vomps (TAAs) adhere to ECM (fibronectin/collagen) + endothelium
| + VirB/D4 T4SS translocates Bep effectors
| + BafA autotransporter binds VEGFR-2 (VEGF analog)
| results in
HIF-1 activation → VEGF secretion (autocrine/paracrine loop)
inhibition of endothelial apoptosis; invasome-mediated uptake
| leads to
ENDOTHELIAL PROLIFERATION → tube formation → ANGIOGENESIS
| results in
Bacillary angiomatosis (skin, LYTIC BONE lesions),
vasoproliferative tumors

Upstream vs downstream. The **initiating lesion** is environmental/behavioral (louse infestation + poverty), not genetic. The **most upstream molecular event** in host colonization is Trw-mediated erythrocyte tropism, which sustains the bacteremic reservoir. The **vasoproliferative arm** is downstream of endothelial adhesion (Vomps/BadA) and effector delivery (VirB/D4-Bep, BafA), all converging on the **HIF-1 → VEGF** hub. Note that Trw

(erythrocyte invasion) and VirB/D4 (endothelial reprogramming) are distinct T4SSs with distinct target cells. The HIF-1/VEGF steps are directly demonstrated for *B. henselae* BadA and **inferred** for *B. quintana* via its homologous Vombs and functional BafA homolog.

Cell types and processes (ontology suggestions). - Erythrocyte (CL:0000232) — intracellular niche; GO:0007155 cell adhesion; GO:0044409 entry into host. - Endothelial cell (CL:0000115) — angiogenesis target; GO:0001525 angiogenesis; GO:0043066 negative regulation of apoptotic process; GO:0001666 response to hypoxia (HIF-1). - Anatomical sites (UBERON): blood (UBERON:0000178), skin/dermis (UBERON:0002067), bone (UBERON:0001474), heart valve/endocardium (UBERON:0002165), lymph node (UBERON:0000029), liver (peliosis, mainly *B. henselae*; UBERON:0002107). - Chemical/biomarker entities (CHEBI): doxycycline (CHEBI:50845); gentamicin (CHEBI:27412).

Comparative species pathophysiology

Feature	<i>B. quintana</i> (trench fever)	<i>B. henselae</i> (cat-scratch)
Reservoir / vector	Human / body louse	Cat / cat flea
Adhesin (TAA)	Vombs	BadA
Risk factors	Homelessness, poverty, lice	Cat contact, flea exposure
BA tissue tropism	Subcutaneous + lytic bone lesions	Peliosis hepatis
Genome	~1.58 Mb (most reduced)	~1.9 Mb

Section-by-Section Template Coverage

- **§1 Disease Info / §5 Infectious agent:** *B. quintana* (α -proteobacterium); MONDO:0005991, ICD-10 A79.0, ICD-11 1C30.0, MeSH D014205. Synonyms: five-day/quintan/shin-bone/wolhynia fever, His-Werner disease, urban trench fever. Aggregated disease-level knowledge.
- **§2 Etiology / risk factors:** Infectious etiology; risk = homelessness, poverty, low income, body-louse infestation, poor hygiene, cold climate/crowding, alcoholism; immunosuppression (HIV) for vasoproliferative disease. **No genetic causal or protective factors** (not heritable); no meaningful gene-environment interaction reported.

- **§3 Phenotypes:** Fever HP:0001945 (relapsing, ~5-day periodicity), Headache HP:0002315, Bone pain HP:0002653 (shin/tibia), Splenomegaly HP:0001744, Bacteremia HP:0031864, Endocarditis HP:0100584, Lymphadenopathy HP:0002716. Onset acute (adult, occupational/social exposure); severity variable (mild self-limited → severe endocarditis).
- **§4 Genetic/molecular: Not applicable to the human host** — no causal genes, pathogenic variants, modifier genes, or chromosomal abnormalities. Relevant "genes" are *bacterial* virulence loci: *trw* (T4SS), *virB/virD4* (T4SS + *bep* effectors), *vomp* family, *bafA*.
- **§6 Mechanism:** See causal chain above.
- **§7 Anatomy:** Blood/erythrocytes (primary), vascular endothelium, skin/dermis, bone (lytic lesions), cardiac valves/endocardium, lymph nodes, spleen.
- **§8 Temporal:** Incubation ~15–30 days; acute relapsing fever often self-limited over weeks; chronic bacteremia/endocarditis can persist months–years if untreated (chronic, insidious course).
- **§9 Epidemiology:** ~5% bacteremia prevalence in sheltered homeless (rising); worldwide wherever body lice occur; predominantly adult; male predominance in homeless cohorts (demographic, not biological). Not inherited — inheritance/penetrance/expressivity sections not applicable.
- **§10 Diagnostics:** Serology (IFA IgG, titers $\geq 1:800$ support endocarditis), prolonged blood culture, PCR/sequencing of *gltA*, *ftsZ*, 16S rRNA; histopathology (Warthin-Starry silver stain) for bacillary angiomatosis; echocardiography for endocarditis. Differential: culture-negative endocarditis (*Coxiella*, *Tropheryma*), other *Bartonella*, epidemic typhus, relapsing fever.
- **§11 Prognosis:** Acute trench fever generally self-limited and non-fatal; endocarditis carries substantial morbidity/mortality and often needs valve surgery; good response to appropriate antibiotics.
- **§12 Treatment:** Doxycycline + gentamicin (chronic bacteremia, RCT-supported); doxycycline $\pm$ aminoglycoside for endocarditis (+ surgery); macrolides/doxycycline for bacillary angiomatosis.
- **§13 Prevention:** Body-lice control (delousing, hygiene, insecticide-treated clothing), improved housing/social conditions, health education; no vaccine ([PMID: 20822446](#)).
- **§14 Other species:** Human reservoir (NCBI:txid9606); molecular detection in cats/dogs/other animals; dogs = natural endocarditis model.
- **§15 Model organisms:** Surrogate rodent-adapted *Bartonella* (*B. birtlesii*, *B. tribocorum*); body louse as vector model; endothelial-cell and erythrocyte in-vitro assays.

Evidence Base

PMID	Title (abbrev.)	Supports finding(s)	Contribution
16494745	<i>B. quintana</i> characteristics and clinical management	F1, F2	Causal agent, host/vector restriction, 1915 origin, clinical spectrum, diagnostics
15643300	<i>Ectoparasitism... 930 homeless, Marseilles</i>	F1, F11	5.3% bacteremia; rising 3.4%→8.4%; 22% louse prevalence
20548954	<i>Trw T4SS mediates host-specific RBC adhesion</i>	F3, F9	Trw T4SS = erythrocyte invasion; surrogate species models
32678094	<i>BafA activates host VEGF pathway</i>	F3	BafA = VEGF analog binding VEGFR-2; <i>B. quintana</i> homolog functional
15891120	<i>High prevalence of B. quintana endocarditis, Sfax</i>	F2	Endocarditis burden (9.8% of all endocarditis)
23602630	<i>Treatment outcomes of human bartonellosis (meta-analysis)</i>	F4	Doxy+gentamicin ↑ resolution of chronic bacteremia
12821469	<i>Randomized trial gentamicin+doxycycline</i>	F10	RCT: 7/7 vs 2/9 eradication (P=0.003)
19696500	<i>Genomics of host-restricted Bartonella</i>	F5	Massive secondary genome reduction; T4SS host adaptation
9407154	<i>Molecular epidemiology of bacillary angiomatosis</i>	F6	Skin/bone tropism; low income/homeless/lice associations
35803580	<i>Vertical transmission in body lice</i>	F7	Horizontal fecal transmission; no transovarial spread
21536788	<i>TAA-dependent adherence under flow</i>	F8	Vomps = <i>B. quintana</i> TAA; ECM/endothelial adhesion
23163798	<i>BadA interferes with VirB/D4 effector translocation</i>	F8	VirB/D4-Bep proangiogenic, anti-apoptotic reprogramming
18627378	<i>BadA head crucial for host cell interaction</i>	F8	TAA adhesion → HIF-1 activation

PMID	Title (abbrev.)	Supports finding(s)	Contribution
38199070	<i>B. quintana</i> in pet cats, Iran	F9	15% cat detection — expanding host range
12860639	<i>Persistent Bartonella bacteremia in humans/animals</i>	F9	Dogs as natural endocarditis model
9986818	<i>Body lice as tools for surveillance</i>	F11	Worldwide distribution in lice
15891141	<i>Autochthonous epidemic typhus + B. quintana</i>	F11	Co-circulation of louse-borne pathogens
20822446	<i>Arthropod-borne diseases & social disorder</i>	F4 (prevention)	Vector control as primary prevention
11871479	<i>Infections in the homeless</i>	F1, F6	Homeless disease context
8727894	<i>B. quintana</i> in seronegative hemodialyzed patient	F2	Chronic adenomegaly; seronegative presentation; gentamicin response

Limitations and Knowledge Gaps

- 1. Mechanistic inference across species.** Much of the vasoproliferative signaling detail (HIF-1 activation, VirB/D4-Bep effector biology, BadA structure–function) is demonstrated in *B. henselae* and **inferred for *B. quintana*** via the homologous Vomps/BafA systems. Direct *B. quintana* endothelial-reprogramming studies are comparatively sparse. The BafA VEGFR-2 interaction was shown functional for the *B. quintana* homolog, but the full downstream cascade in *B. quintana* is extrapolated.
- 2. Treatment evidence is thin.** The pivotal RCT enrolled only 20 patients ([PMID: 12821469](#)), and the meta-analysis explicitly flags that most treatment evidence is observational and at high risk of bias ([PMID: 23602630](#)). Optimal duration for endocarditis, and management of relapse, remain guideline/expert-opinion driven.

3. **Epidemiology is likely underestimated.** Fastidious culture requirements and non-specific clinical presentation mean cases are under-ascertained; prevalence figures derive largely from urban homeless cohorts in France and may not generalize globally. Population-level incidence rates (new cases/100,000/yr) are not well established.
4. **Expanding host range uncertain in significance.** Detection of *B. quintana* DNA in cats, dogs, non-human primates, and other animals ([PMID: 38199070](#)) raises the question of whether animals are competent reservoirs or incidental/spillover hosts. Whether these represent transmission risk to humans is unresolved.
5. **No human genetic component** — the template's genetic/variant/inheritance/heritable-model sections are **not applicable**; any "susceptibility" is socioeconomic and behavioral, not genomic.
6. **Quality-of-life and disability data** specific to trench fever (EQ-5D, SF-36, PROMIS) are essentially absent in the literature; QoL impact is inferred from the burden of chronic bacteremia/endocarditis and the comorbidities of the affected homeless population.

Proposed Follow-up Experiments / Actions

1. **Direct *B. quintana* vasoproliferation assays.** Test purified *B. quintana* BafA and Vombs in human endothelial cells for HIF-1 stabilization, VEGF secretion, tube formation, and apoptosis inhibition — to replace *B. henselae* inference with species-specific data.
2. **Larger, multicenter treatment trial.** A pragmatic RCT (or well-designed prospective registry) across multiple homeless-service settings to define the optimal antibiotic regimen and duration for chronic bacteremia and endocarditis, and to quantify relapse.
3. **Global surveillance via body lice.** Extend the molecular-surveillance approach ([PMID: 9986818](#)) to systematically map *B. quintana* prevalence in body lice and homeless populations worldwide, including incidence estimation.
4. **Reservoir-competence studies in animals.** Determine whether cats/dogs/primates with detectable *B. quintana* DNA harbor viable, transmissible bacteria and whether alternative arthropod vectors participate — clarifying zoonotic risk.
5. **Public-health intervention evaluation.** Rigorously evaluate delousing/hygiene/housing interventions for their effect on *B. quintana* incidence, integrating vector control per [PMID: 20822446](#).

6. **Biomarker/diagnostic development.** Develop rapid point-of-care serologic or molecular assays suited to low-resource, homeless-service settings to improve case ascertainment and shorten time-to-treatment.

Consensus Answer

Trench fever is a **louse-borne bacterial infection caused by *Bartonella quintana*** (MONDO:0005991; ICD-10 A79.0), a human-restricted α -proteobacterium transmitted through the feces of the body louse (*Pediculus humanus humanus*) and re-emerging among the urban homeless worldwide. It is **not a genetic disease** — risk is socioeconomic (homelessness, poverty, body lice, poor hygiene; immunosuppression for vasoproliferative forms) — and it spans a clinical spectrum from classic relapsing five-day fever with shin pain to chronic bacteremia, culture-negative endocarditis, and bacillary angiomatosis, driven mechanistically by **Trw-T4SS-mediated erythrocyte invasion** plus **Vomps/BadA-, VirB/D4-Bep-, and BafA-driven HIF-1/VEGF endothelial proliferation**. It is effectively treated with **doxycycline plus gentamicin** (randomized-trial evidence) and prevented by **body-louse control**; no vaccine exists.

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